

The Center Is Half

The Zero-Mean Tilt as the Mirror in Tilt Space, from $P(+W)/P(-W) = e^W$

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

The detailed mirror $P(+W)/P(-W) = e^W$ of Ch.77 has now been read in two compressions: Ch.78 in cumulants ($k_{-n} = (-1)^n k_n$) and Ch.79 in the rate function ($l(a) = l(2\mu - a)$). This chapter reads the mirror in the TILT ITSELF. The tilt is how the tail is actually measured (the biased ensemble); its mean is $a(t) = \langle W e^{(-tW)} \rangle / \langle e^{(-tW)} \rangle = -F'(t)$, the derivative of the CGF $F(t) = \ln \langle e^{(-tW)} \rangle$. The exchange symmetry $F(t) = F(1-t)$ (a direct corollary of $P(w) = e^w P(-w)$) forces $a(t) = -a(1-t)$: the tilted mean is ODD about $t = 1/2$, so the zero-mean tilt $a(1/2) = 0$ is the mirror's center, an EXACT statement on every reversible cycle - Gaussian or not, $q = 0.90$ or not. Measured: the $q = 0.90$ control obeys $a(t) + a(1-t) = 0$ to $7e-4$ across the whole grid with $a(1/2) = -0.0004$. The feedback coin does NOT bend this curve - it SHIFTS it: $a(t) + a(1-t)$ is a constant -0.145 engaged and -0.258 harvest at EVERY t , i.e. the center $a(1/2)$ moves to -0.073 and -0.129 by exactly the coin, all antisymmetry preserved about the shifted center. One price, paid as a center.

1. The Tilt Is How the Tail Is Measured

The Legendre dual of Ch.79, $l(a) = \sup_t [at - F(t)]$, is conjugate to sweeping the tilt (Touchette 2009, the biased/tilted ensemble). Reweight by $e^{(-tW)}$; the mean becomes

$$a(t) = \langle W e^{(-tW)} \rangle / \langle e^{(-tW)} \rangle = -F'(t)$$

and the rate function at the point $a(t)$ is read by the identity $l(a(t)) = a(t)t - F(t)$ at every t . For the Gaussian parabola ($\mu = +5.98$, $\sigma^2 = 12.04$) this gives $a(t) = \mu - \sigma^2 t$, linear, and a perfectly quadratic $l(a(t)) = \sigma^2 t^2/2$ - the Ch.79 curve traced by one coordinate. The tail is not guessed; it is SWEPT.

2. The Mirror in Tilt Space: $a(1/2) = 0$

Because $F(t) = F(1-t)$ is even about $1/2$ (a direct corollary of the detailed theorem, as in Ch.78), its derivative is odd about $1/2$, so

$$a(1/2) = 0 \text{ exactly, on EVERY reversible } \Delta F = 0 \text{ cycle; } a(t) = -a(1-t)$$

The zero-mean tilt - the tilt that makes the net work vanish on average - is ALWAYS $1/2$, independent of the shape of P : this is the whole-curve form of the mirror, strictly stronger than any single cumulant of Ch.78. Measured on the $q = 0.90$ control (250k):

t	$a(t)$	$a(1-t)$	$a(t)+a(1-t)$
0.00	+0.1131	-0.1130	+0.0001
0.25	+0.0542	-0.0547	-0.0006
0.40	+0.0212	-0.0219	-0.0007
0.50	-0.0004	-0.0004	-0.0007

A non-Gaussian (skew $+1.30$, kurt 7.0 , $q = 0.90$) mirror, exact to $7e-4$ across the grid. The detailed theorem fixes the whole curve of tilted means, not just the ratios.

3. The Coin Moves the Center, Not the Mirror

Feedback breaks the plain mirror - but in tilt space its signature is a SHIFT, not a bend:

row	$a(1/2)$	$a(t)+a(1-t)$ (constant, all t)
control	-0.0004	~ 0
engaged	-0.0726	-0.145
harvest	-0.1294	-0.258

The antisymmetry $a(t) + a(1-t) = \text{const}$ holds at EVERY t for the coin rows - the shape is untouched, but the center $a(1/2)$ is displaced by exactly half of that constant (-0.073 engaged, -0.129 harvest). The demon does not warp the mirror; the mirror is translated by the bit. One coin, paid as a center of the tilt curve, the same single entry that priced knowing in variance (Ch.76), cumulants (Ch.78), and the whole tail (Ch.79).

4. The 0/0 in Tilt Space

Where is the 0/0? At the mirror center $a(1/2) = 0$ the ratio $P(w)/P(-w)$ at $w = 0$ is the Ch.77 0/0 removed to 1 - and the tilted mean there is the slope $P'(0)/P(0)$. On the parabola the center is reached at $t^* = \mu/\sigma^2 = 0.496 \sim 1/2$ (the Gaussian mode reflected through the mirror); the 0/0 of the tail (empty higher cumulants, Ch.78) is the statement that $a(t)$ is EXACTLY linear and $V(t)$ constant, so no curvature is left to mirror. Every reversible cycle carries the same center at $1/2$; the coin is the only thing that moves it.

Main Results

THEOREM

The Center Is Half: on every $\Delta F = 0$, time-reversible round trip, the tilted mean $a(t) = -F'(t)$ is odd about $t = 1/2$, so the zero-mean tilt satisfies $a(1/2) = 0$ exactly, whatever the shape of P - measured $a(t)+a(1-t) = 0$ to $7e-4$ on the $q = 0.90$ control.

COROLLARY

The Coin Moves the Center, Not the Mirror: feedback displaces the center to $a(1/2) = -0.073$ engaged and -0.129 harvest while preserving the antisymmetry about the shifted center ($a(t)+a(1-t)$ constant at every t) - one coin, paid as a pure translation of the tilt curve.

COROLLARY

The Tilt Traces the Mirror: sweeping t recovers the Ch.79 rate function by the Legendre identity $l(a(t)) = a(t)t - F(t)$, exact for the Gaussian parabola (a linear, V constant) and everywhere the mirror is read at whole-curve strength.

Conclusion

Ch.77 measured the mirror in the PDF, Ch.78 in the cumulants, Ch.79 in the rate function; this chapter reads it where the tail is actually measured - in the tilt itself. The zero-mean tilt is the mirror's center, and it is exactly 1/2 on every reversible cycle, Gaussian or not. The coin, which bent every ledger in earlier chapters, here only translates the center - the mirror moves by the bit, and the shape is preserved. Six lenses on one equality; the demon pays the same single coin in every one.

References

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- [2] Jarzynski, C. (1997). Phys. Rev. Lett. 78, 2690.
- [3] Touchette, H. (2009). Phys. Rep. 478, 1 (tilted/biased ensemble, Legendre duality).
- [4] Ellis, R. (1985) / Gartner-Ellis theorem (rate functions).
- [5] Kitahara-Imada (2013) tilted ensembles for Brownian motors.
- [6] Book chain: Ch.71 (integral FT), Ch.77 (detailed mirror), Ch.78 (cumulant ladder), Ch.79 (rate function), Ch.76 (TUR).